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Business Process Modeling

Project
Car Repair

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Introduction

In this project, we modeled and analyzed the processes between the customer and the repair shop. The process begins with the customer contacting the repair shop. During the contact, the customer can choose between two options: either request a car pick up from the destination or ask for an appointment to bring the car to the shop.

If the customer chooses to pick up the car, he/she can send the location of the car or cancel the request. If the customer chooses to drive the car to the shop, the shop offers them a date and time, and the customer can either accept this offer, offer a different date/time, or cancel the request. When the customer offers a different date/time, the shop can either confirm this offer or offer a different time. This process of mutual offers continues until an appointment is fixed or the request is canceled.

After the appointment is accepted, the customer can additionally ask for a courtesy car. The repair shop can then offer different car options and prices. The customer can accept these offers, reject them, or cancel the request for repairs altogether. All these processes end when the car is delivered to the workshop or the request is canceled.

Additionally, based on the requirements specified in the project brief, a design change was also implemented in the model to allow the process to restart after the request is canceled so that the customer can contact another shop, and this change was analyzed.

Initially, the process was modeled at a more abstract level using BPMN notation, and then this model was transformed into a workflow net (WF-net) for formal analysis.

Abstract level Modeling

The business process was modeled using BPMN via the BPMN.io online editor. The diagram (Figure 1) consists of two main pools representing the two actors involved: Customer and Repair Shop.

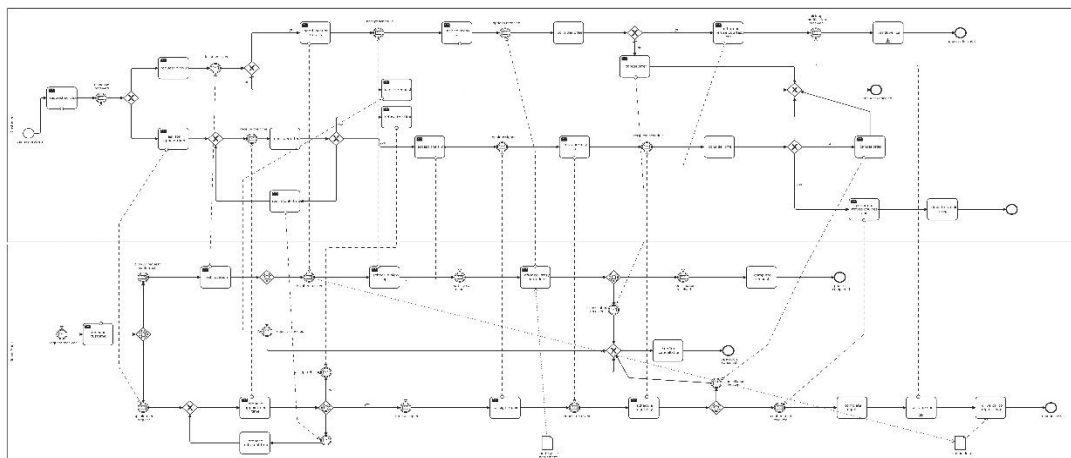


Figure 1: BPMN diagram of the processes

Message flows illustrate the communication between participants, connecting tasks and events across the pools. The model uses flat pools without lanes to simplify the representation of each participant's activities.

Each pool ends with three possible outcomes, modeled as separate end events: successful delivery of the car, a partial process ending, or cancellation of the request. Only exclusive gateways were used to handle decision points, and inclusive gateways were avoided to ensure that the model could be accurately converted into a Petri net.

The process begins on the customer's side when they initiate contact with the shop. After that, a decision is made between two options: requesting a pick-up for the car or asking for an appointment to drive it to the shop. Each choice triggers its own message exchange flow with the shop.

Two important loops are present in the model. The first (*Figure 2*) appears during appointment scheduling: the shop proposes a time, and the customer may either accept, reject and suggest a new time, or cancel the request. This loop continues until one of the parties agrees or cancels the process. The second loop (*Figure 3*) involves courtesy car request, where multiple rounds of offer and response may occur before a decision is made.

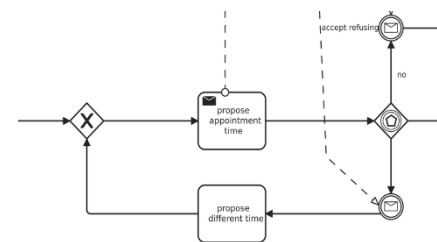


Figure 2: repair shop's loop

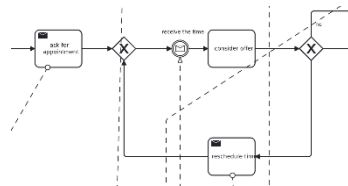


Figure 3: customer's loop

In response to the project requirement, the model includes a redesign: when the customer cancels the process, instead of ending the interaction, they initiate a search for another repair shop, represented as a sub-process. After that, the flow returns to the initial decision point, allowing the customer to restart the process with a new provider.

Workflow Net Analysis

Before analyzing the behavior of the process, the BPMN model was transformed into a Petri net structured as a Workflow Net (WF-net). This transformation ensures that the process starts from a unique initial place and can always reach a unique final place, meeting the structural requirements of a WF-net.

To perform the transformation correctly, the BPMN model was slightly simplified. No inclusive gateways were used, which was beneficial since such elements cannot be directly represented in Petri nets. The model was built entirely with exclusive gateways, making it easier to convert decision points into Petri net elements. Sub-processes in the BPMN model were also minimized.

In our case, sub-processes such as the search for another repair shop were modeled as a single transition, as their internal details would not affect the correct analysis.

The resulting Petri net contains a single-entry point and one final sink place. All transitions are part of a path that connects the start to the end. The car pick-up, appointment scheduling, and courtesy car negotiation stages are each represented using corresponding Petri net structures—including choices, message synchronizations, and loops.

The loops in the model, particularly in the appointment and courtesy car stages, were modeled by connecting XOR-split and XOR-join patterns (Figure 4 and 5 respectively). This allowed back-and-forth interaction between the customer and the shop to be accurately captured. The customer module originally contained three end events, and these were unified into one common final place during the transformation, as required by the definition of WF-nets.

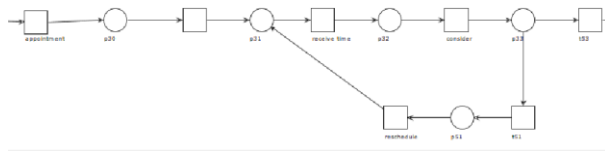


Figure 4: appointment loop of customer

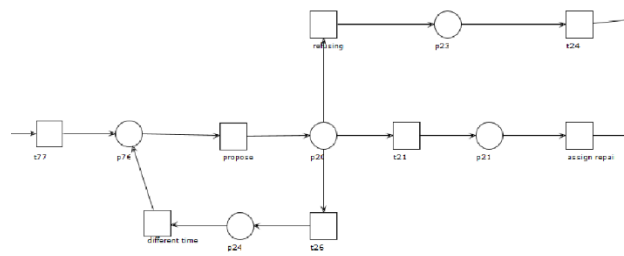


Figure 5: courtesy car loop of repair shop

Each module (customer and repair shop) was first analyzed separately. During this phase, the input and output interface places were not considered, to evaluate the internal soundness of the components. Both modules were found to be structurally sound: they had no unreachable transitions or deadlocks, and each transition belonged to at least one execution path leading to the final marking.

The integrated workflow net-obtained by connecting the two modules based on message flows was analyzed using workflow analysis tools. Waiting points and synchronization between actors were implemented using intermediate places and transitions. A connector transition was also added to unify the final places of the individual modules into a shared final state.

From a structural perspective, the resulting net satisfies the key workflow properties. The net is bounded (1-safe), meaning no place ever contains more than one token. Its reachability graph was finite and showed no deadlocks. All transitions could be triggered under at least one execution scenario, confirming that the net is live and sound.

The overall net was also checked for additional properties:

- the net is not fully free-choice, as analysis reported several violations due to overlapping input conditions in decision points;

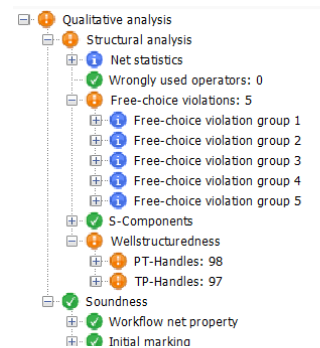


Figure 6: Analysis of complete graph

- the net contains a single S-component, covering all places and transitions, which indicates structural cohesion;
- the model is not well-structured, as it includes 98 PT-handles and 97 TP-handles, mostly around synchronization points and the re-entry loop;
- the presence of positive place- and transition-invariants confirmed that the token flow is conserved and cycles are correctly formed.

Based on the current design, the full workflow net contains approximately 83 transitions and 93 places, consistent with the combined structure of the customer and shop modules. The estimated reachability graph has a moderate size and includes no deadlocks, confirming that the model behaves correctly under all execution paths.

The Petri net was analyzed using the Woflan plugin in ProM, (Figure 7) which confirmed that the model is a sound workflow net.

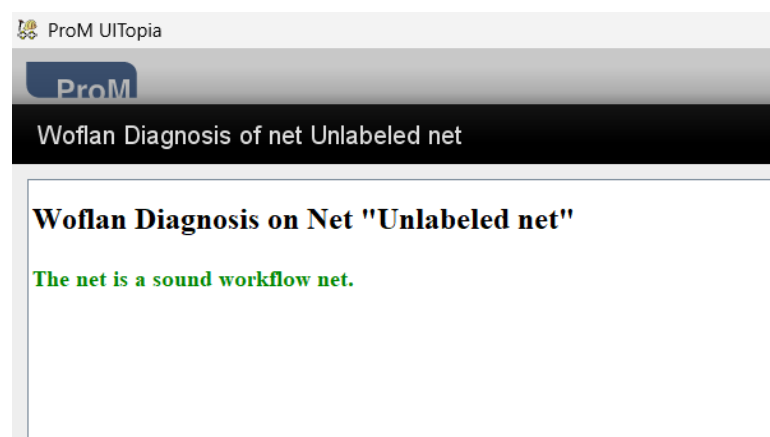
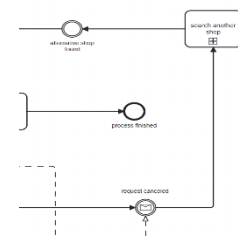


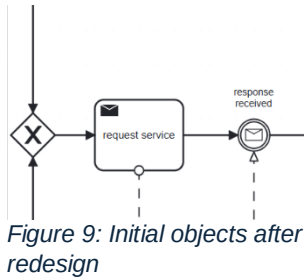
Figure 7: Sound workflow net validation result from ProM

Redesign Analysis

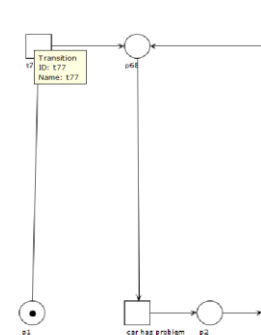
In line with the extended requirements of the project, the original process was modified to allow the customer to restart the interaction after canceling a repair request. This redesign ensures that if a customer is not satisfied with the ongoing negotiation or services, they are able to search for a different repair shop and initiate the process again without fully terminating the overall workflow.

In the BPMN model, this was implemented by connecting the cancellation path to a sub-process named “Search Another Shop”. (Figure 8) After this sub-process, the customer flow returns to the initial decision point, where a new repair request can be started. The loop enables multiple interactions with potentially different providers, all within the same process model. (Figure 9)





This redesign directly affects the structure of the Petri net. Specifically, the WF-net now includes a backward arc from the final place back to the initial place, turning the original acyclic net into a cyclic one. (Figure 10) While this allows process continuity, it technically violates the classical definition of a workflow net, which assumes a single termination point.



As a result, the redesigned net is no longer strictly sound under the formal definition, since there is no guaranteed final marking where the process ends once and for all. The net remains bounded and safe, as only one token exists at any time, and each place holds at most one token. However, it now exhibits relaxed soundness, meaning that each subprocess (or cycle) behaves correctly and leads to a valid continuation of the process, even though the global model has no absolute end. (Figure 11) This change introduces free-choice violations due to the loop returning to an already marked place, causing nondeterminism in process resumption. Despite these limitations, the redesign captures the real-life scenario where customers often retry with different providers and thus improves the model's expressiveness.

Conclusion

The modeling and analysis of the car repair scenario showed that even with multiple outcomes and interactions between participants, it is possible to design a process that behaves correctly. Using BPMN, the main flow was captured with a clear separation of roles, and the transformation into a workflow net allowed formal validation. An additional loop was introduced to let the customer restart the process after canceling, adding flexibility. Despite this, the final Petri net remained sound, as confirmed by WoPeD. While describing the complete network, message flows between participants causes structural flaws such as free-choice conflicts and handles. However, they do not affect the model's ability to execute and complete correctly.

